

WAIOURU, New Zealand

WMO: 933390

Lat: 39.446S

Lon: 175.658E

Elev: 2687

StdP: 13.32

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.0	27.2	21.6	17.9	26.9	24.1	20.1	28.9	34.5	44.1	27.3	44.8	3.2	100	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.9	72.3	60.8	70.0	59.6	67.9	58.7	63.2	69.1	61.6	67.3	60.1	65.7	8.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.9	88.2	66.4	59.2	83.0	64.6	57.6	78.2	63.3	30.1	69.3	28.8	67.4	27.7	65.8	69.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	20.2	17.4	20.6	77.4	1.7	2.8	19.3	79.5	18.3	81.1	17.4	82.7	16.1	84.7
			19.6	66.2	1.8	1.6	18.3	67.3	17.3	68.3	16.3	69.2	15.0	70.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	56.9	57.4	53.9	49.0	45.0	40.5	39.3	40.2	42.9	45.8	49.1	54.1	
	DBStd	7.98	5.25	4.89	5.03	5.09	5.38	5.14	4.54	4.35	4.65	4.79	5.10	4.99	
	HDD50	1672	10	5	23	78	172	288	333	304	216	145	77	21	
	HDD65	6295	255	216	345	481	620	736	798	769	662	596	478	339	
	CDD50	857	222	211	143	47	17	2	0	0	4	14	49	148	
	CDD65	4	2	2	0	0	0	0	0	0	0	0	0	0	
	CDH74	32	18	11	1	0	0	0	0	0	0	0	0	2	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	6.7	6.6	6.0	5.8	5.6	6.1	6.5	6.6	6.8	7.6	7.8	7.7	7.3	
Precipitation	PrecAvg	78.1	5.4	4.3	5.2	5.7	6.7	8.2	7.9	7.4	7.6	7.1	6.1	6.9	
	PrecMax	94.6	8.9	17.8	9.9	9.9	11.3	12.5	13.9	11.4	18.4	15.1	9.6	10.2	
	PrecMin	63.7	2.2	0.7	2.2	3.5	2.6	4.4	2.8	3.0	1.5	2.8	2.4	3.1	
	PrecStd	9.6	1.4	3.5	2.1	1.9	2.0	2.4	3.0	2.3	3.5	2.4	2.1	1.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.6	75.4	71.5	65.4	61.1	56.4	53.3	54.2	59.2	62.6	69.3	71.6	
		MCWB	61.4	62.6	58.1	56.9	54.7	51.6	48.8	48.0	51.6	53.5	57.2	59.4	
	2%	DB	72.6	72.3	68.3	62.3	58.2	52.7	50.5	51.6	56.0	59.5	65.2	68.4	
		MCWB	61.4	61.2	58.2	55.2	53.2	49.0	46.6	46.2	49.3	51.7	55.5	58.4	
	5%	DB	70.0	69.9	66.1	60.1	55.8	50.6	48.5	49.5	53.7	57.3	62.0	66.1	
		MCWB	59.9	60.1	57.2	54.4	51.6	47.4	44.9	44.9	48.4	50.4	53.9	57.5	
	10%	DB	67.7	67.5	63.7	58.1	53.5	48.7	46.6	47.5	51.6	55.0	59.3	63.9	
		MCWB	58.6	59.3	56.4	53.1	49.8	45.9	43.4	43.8	46.9	49.4	52.4	56.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.0	65.6	62.8	59.7	56.3	52.1	49.5	49.9	52.8	54.8	59.2	62.5
			MCDB	72.7	72.1	67.4	62.8	59.5	55.0	51.9	52.9	56.9	60.7	66.3	68.6
2%		WB	63.4	63.4	60.2	57.0	54.0	49.7	47.2	47.3	50.8	52.8	57.0	60.3	
		MCDB	69.8	69.1	64.9	60.4	57.0	52.1	49.9	50.3	54.6	57.7	62.8	65.6	
5%		WB	61.5	61.8	58.4	55.4	52.0	47.7	45.3	45.6	49.1	51.3	54.9	58.7	
		MCDB	67.7	67.5	63.7	59.0	55.3	50.0	47.9	48.6	52.8	56.0	60.3	64.0	
10%		WB	59.8	60.2	57.1	53.6	50.1	46.1	43.7	44.1	47.4	49.9	53.2	57.3	
		MCDB	65.8	65.9	62.5	57.4	53.1	48.4	46.1	47.0	51.1	54.4	58.5	62.8	
Mean Daily Temperature Range	5% DB	MDBR	18.3	17.9	17.3	15.5	13.0	11.8	11.3	12.5	14.2	15.1	16.6	16.3	
		MCDBR	21.9	20.9	20.7	17.9	15.0	12.6	12.7	15.3	17.1	18.6	20.6	19.3	
		MCWBR	12.7	11.6	12.0	11.9	10.7	9.7	9.6	10.9	12.0	12.4	13.3	11.7	
	5% WB	MCDBR	18.1	16.8	16.3	13.9	12.6	10.8	10.8	12.6	14.5	16.4	18.1	16.4	
		MCWBR	11.4	10.2	10.3	9.9	9.6	9.0	8.7	9.7	10.8	11.9	12.5	10.9	
Clear-Sky Solar Irradiance	taub	0.310	0.306	0.289	0.281	0.266	0.264	0.260	0.270	0.291	0.303	0.298	0.310		
	taud	2.577	2.604	2.663	2.664	2.700	2.675	2.694	2.664	2.566	2.538	2.580	2.578		
	Ebn at Noon	324	318	312	294	281	271	280	294	305	315	326	326		
	Edh at Noon	33	30	26	23	19	18	19	23	29	32	33	33		
All-Sky Solar Radiation	RadAvg	2062	1793	1429	982	656	508	586	811	1138	1487	1831	1909		
	RadStd	155	181	83	65	43	43	49	62	98	104	175	156		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	+1.33	+1.51	+1.15	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	+1.06	N/A	N/A	N/A	-45	N/A	N/A	N/A

Nomenclature: See separate page